Day 30 - Complex Roots

Wednesday, April 2, 2025 9:32 AM

Objective:

- Students will …
  - investigate the number of linear and quadratic factors a polynomial can have
  - visualize complex numbers and complex roots for a quadratic function

Mathematical Practice(s): reason abstractly and quantitatively, attend to precision, look for and make use of structure

Classwork:

- 8-101
- 8-102
- 8-137
- 8-81

Homework:

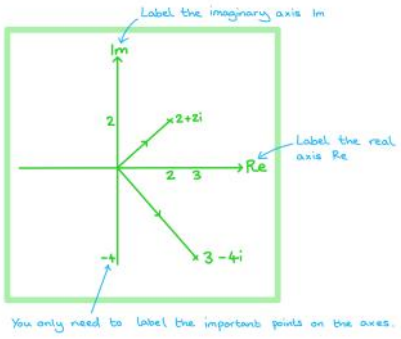
- Day 30 HW - Polynomial Growth Check

Announcement:

- Day 33: Polynomial Team Task
  - ODD - Monday, 4/28
  - EVEN - Tuesday, 4/29
- Day 34: KC #1 - Polynomials
  - ODD - Wednesday, 4/30
  - EVEN - Thursday, 5/1

Key Notes:

Argand Diagram



Day 30 Classwork

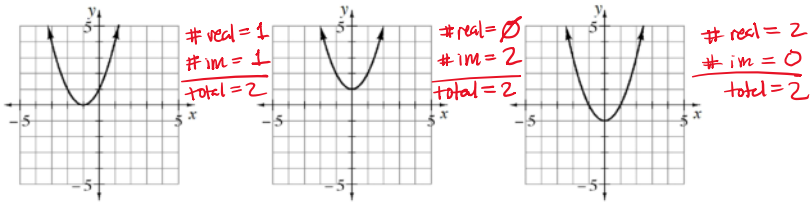
Algebra 2 Honors Name: _____

Day 30 | Complex Numbers Cont.

Warm-Up

Fundamental Thm of Algebra: degree (n) = # real roots + # imaginary roots

8-101. Recall that a polynomial function with degree n crosses the x-axis at most n times. For instance, $y = (x+1)^2$ intersects the x-axis once, while $y = x^2 + 1$ does not intersect it at all. The function $y = x^2 - 1$ intersects it twice. These graphs are shown below.



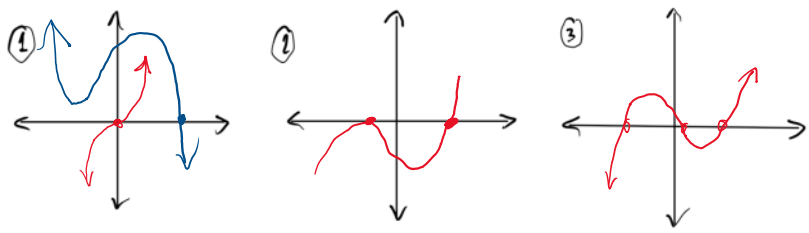
# real = 1
# im = 1
total = 2

# real = ∅
# im = 2
total = 2

# real = 2
# im = 0
total = 2

$y = (x+1)^2$ $y = x^2 + 1$ $y = x^2 - 1$

a. The graph of a third-degree equation might intersect the x-axis one, two, or three times. Make sketches of all these possibilities.



b. Can a third-degree equation have zero real roots? Explain why or why not.

odd = No because of the end behavior, it will eventually have to cross it at least 1 time

even = yes! end behavior same direction

translated above or below the x-axis

8-102. Consider the function $f(x) = x^3 - 3x^2 + 3x - 2$.

a. How many real solutions **could** $f(x)$ have?

b. Given that $f(x) = x^3 - 3x^2 + 3x - 2 = (x - 2)(x^2 - x + 1)$, find all solutions of $f(x)$.

c. How many real roots and how many non-real (complex) roots does $f(x)$ have?

8-81. In parts (a) through (d) below, look for patterns as you calculate the **product** for each pair of complex numbers. Use what you find to answer parts (e) through (g).

a. $(2+i)(2-i)$

$= \boxed{5}$

	2	$-i$
2	4	$\cancel{-2i}$
$+i$	$\cancel{+2i}$	$-i^2$

$= -(-1) = 1$

b. $(3-5i)(3+5i)$

c. $(-4+i)(-4-i)$

d. $(1+i\sqrt{3})(1-i\sqrt{3})$

e. What complex number can you multiply $3+2i$ by to get a real number?

f. What happens when you multiply $(-4+5i)(-4+3i)$?

g. What complex number can you multiply $a+bi$ by to get a real number?